Cell Segmentation for Spatial Transcriptomics

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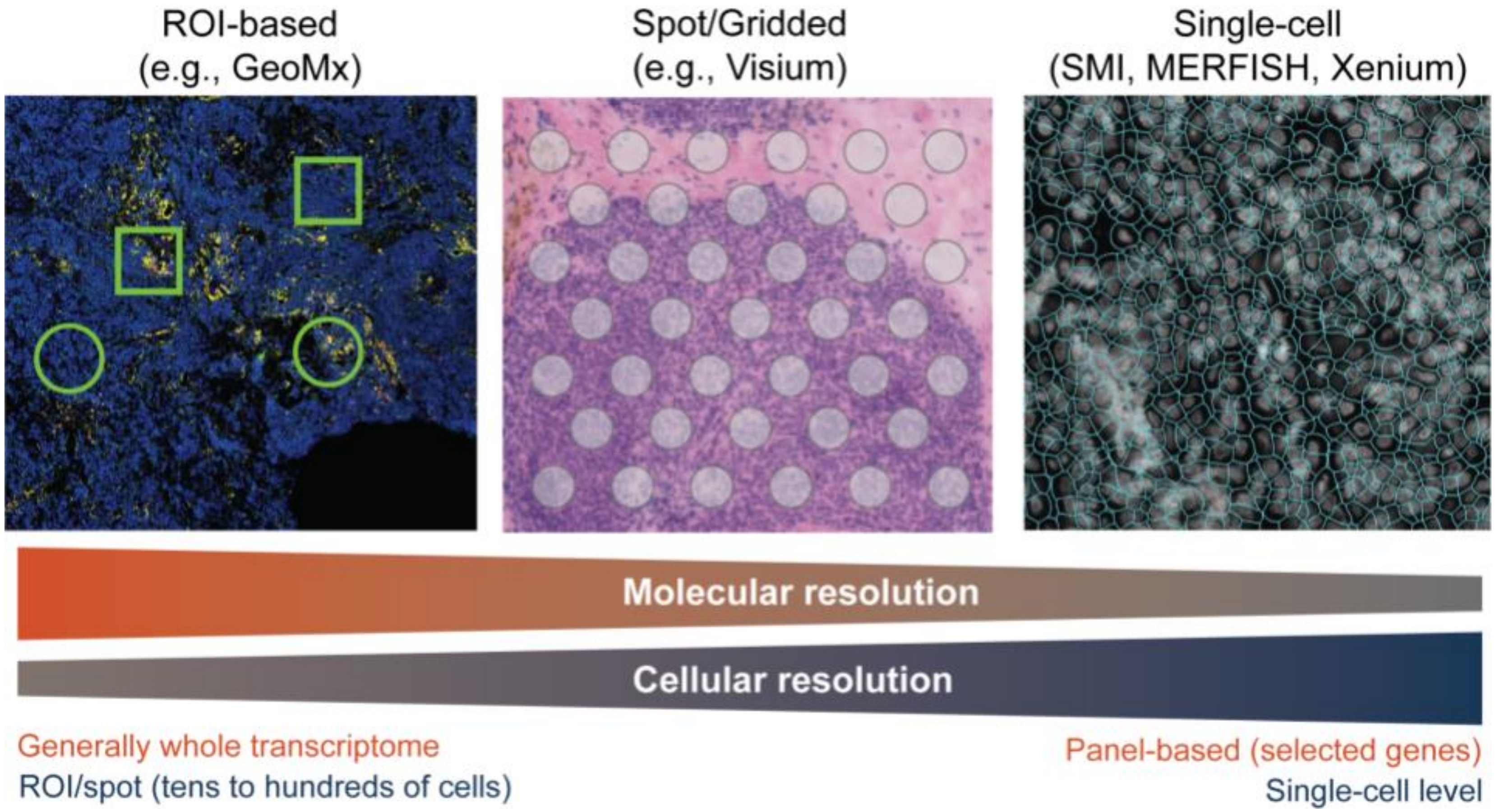
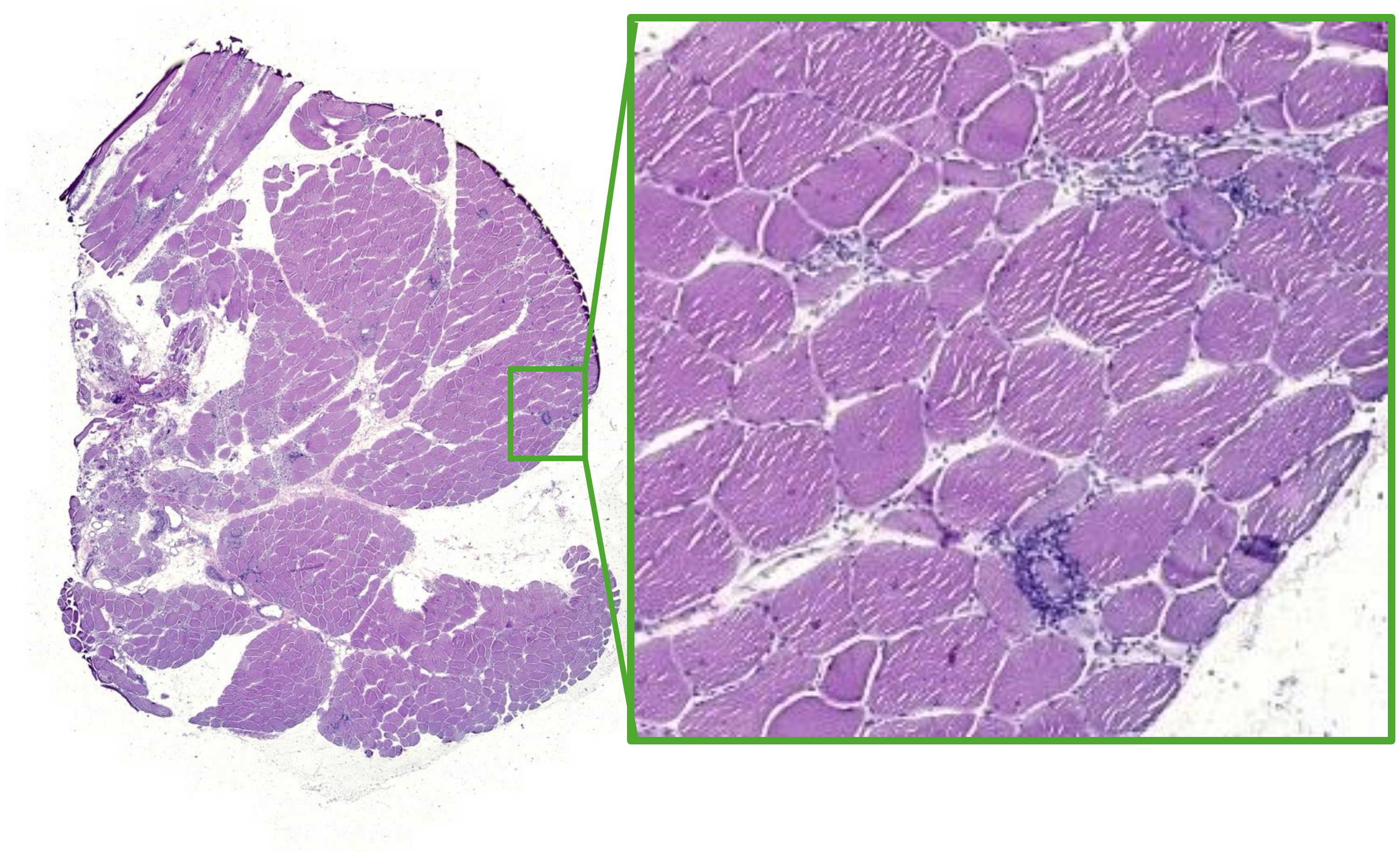
Evolution of Spatial Transcriptomics platforms



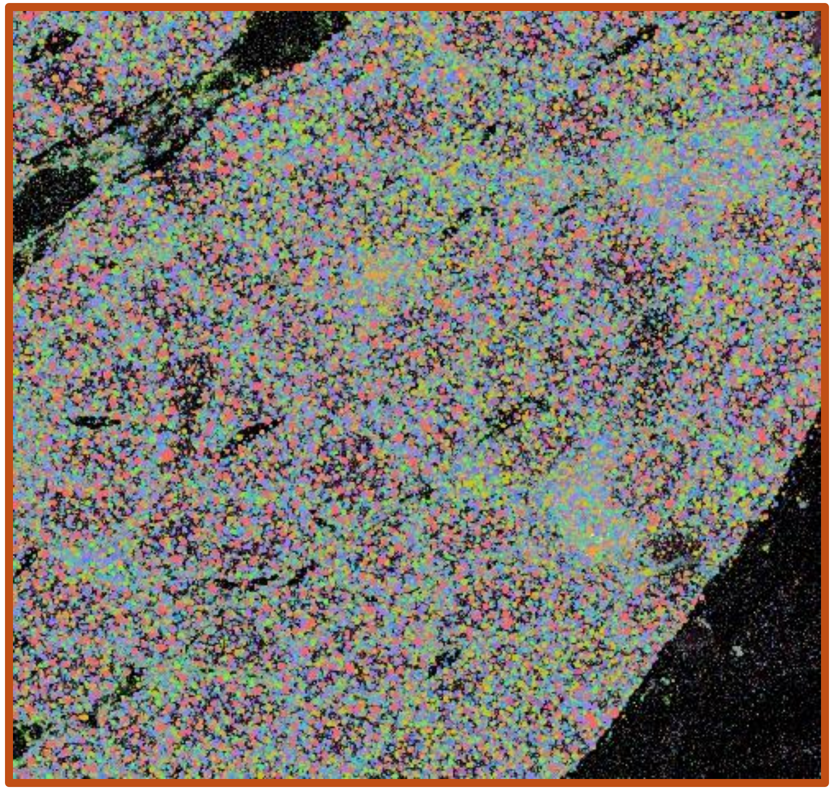
Source: https://hutchdatascience.org/Choosing_Genomics_Tools/spatial-transcriptomics-1.html

From pathology to molecular insights

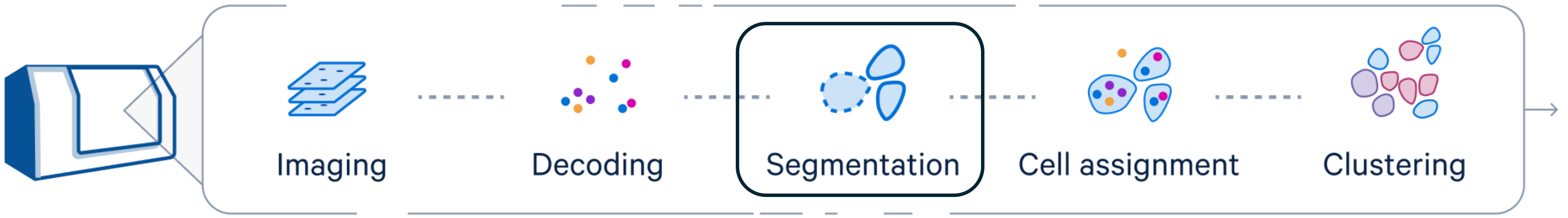
H&E staining of inclusion body myositis muscle tissue



Xenium 480-plex ST dataset



Spatial Transcriptomics workflow



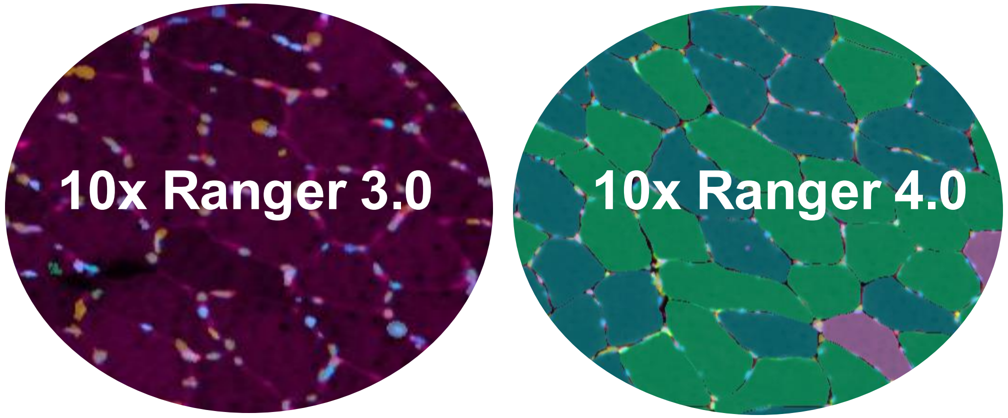
Downstream analysis

Aim: accurately define a ‘cell’ – crucial for all downstream analysis

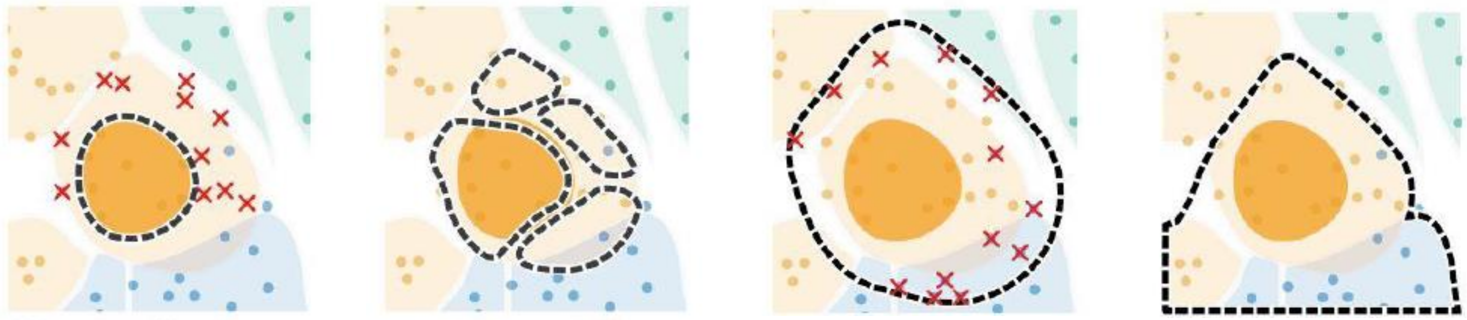
Many platforms provide in-house segmentation support

- CosMx SMI
- MERSCOPE/MERFISH
- 10x Xenium

..but don’t always accurately segment your specific cell type



Challenges Cell Segmentation



False negatives   Over-segmentation   False positives   Under-segmentation

Data limitations depending on platform

- Counter stains (membrane, nuclei, organelles, cell fill)
- Sparse transcript expression
- Unclear ground truth

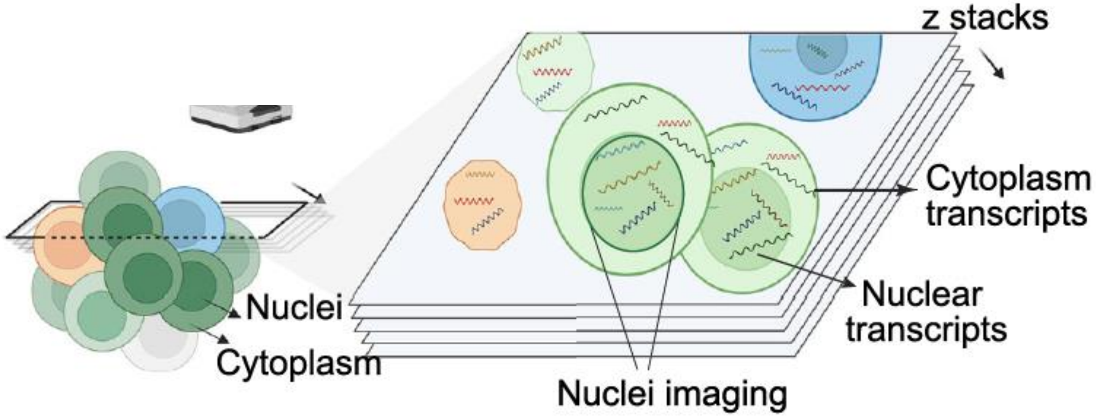
Progress in the field

- Counter stains/ segmentation kits
- Algoritms for cell segmentation

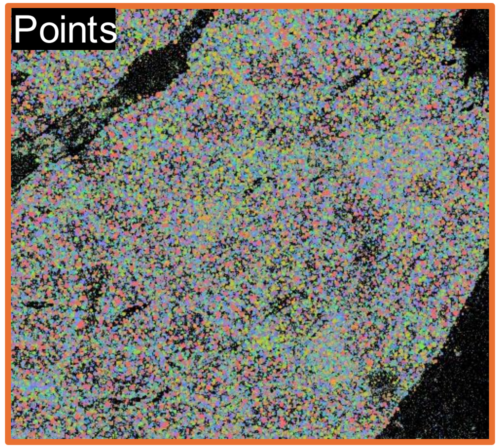
Image Source: https://elihei2.github.io/segger_dev/

Spatial datasets: complex data collection

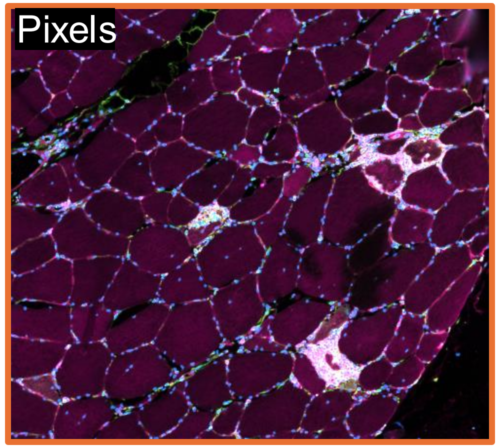
ST dataset



Transcripts (x,y,z)



Images



Segmentation Polygons

Cell boundaries
+
Transcripts

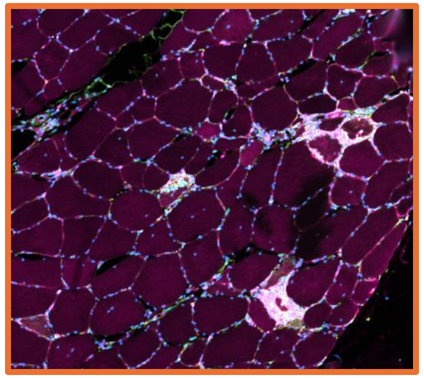
Cell by gene matrix / Anndata table

Downstream analysis

Segmentation methods
- Image-based
- Transcript-based
- Combined methods

Cell Segmention Methods

- **Image-based**



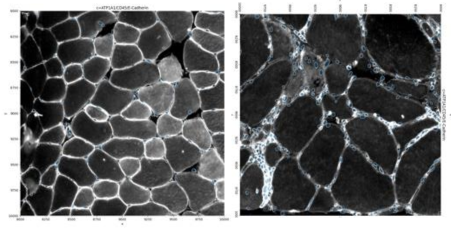
Input: image

Captures morphology; aligns with histology

Sensitive to image quality/staining

- storage of samples
- species
- complicated cell morphology
- disease associated morphology

Mouse Human



Cellpose **Mesmer**

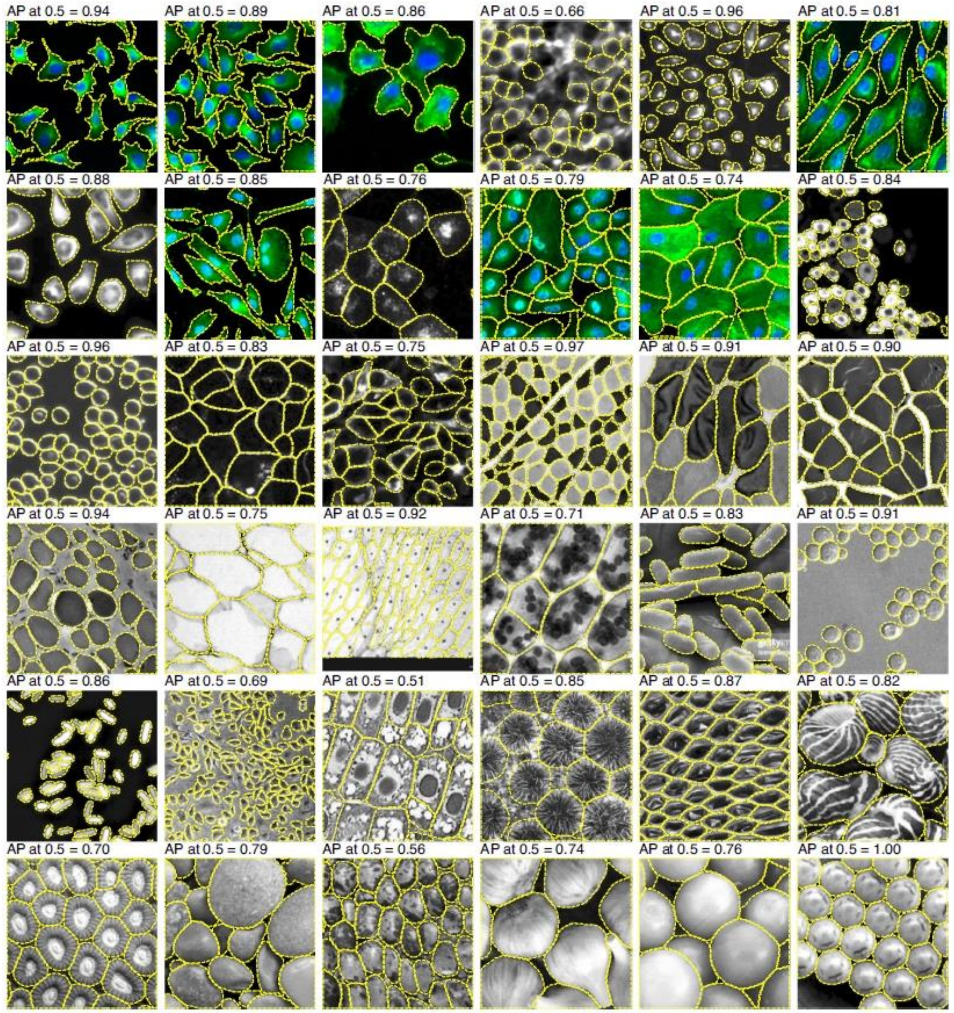
UNSEG

ARTICLES
https://doi.org/10.1038/s41592-020-01018-x
nature methods
Check for updates

Cellpose: a generalist algorithm for cellular segmentation

Carsen Stringer, Tim Wang, Michalis Michaelos and Marius Pachitariu

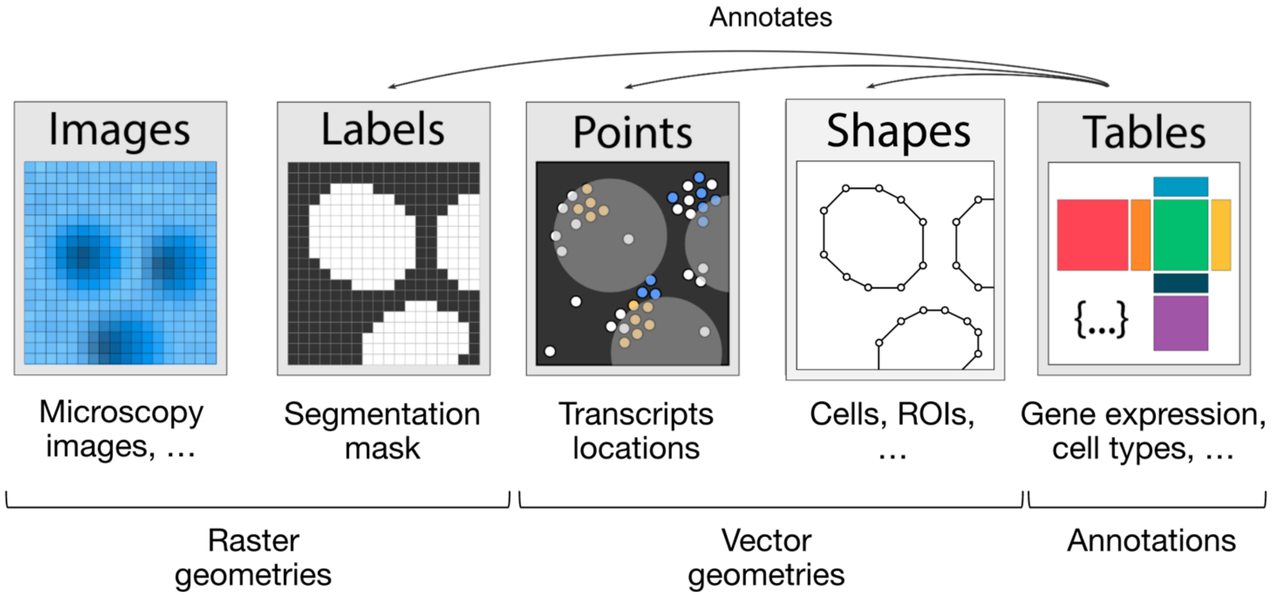
- Generalist, deep-learning based segmentation method
- Different pretrained models to choose from, possibility to train your own model
- Image processing (deblurring, upsampling, contrast enhancement
- Parameter fine-tuning (cell size, circularity)
- Considered the gold standard in image-based segmentation



Cellpose for Spatial Transcriptomics data

- Run it on your morphology image
  associated OME metadata for transcript allocation (practical_1)

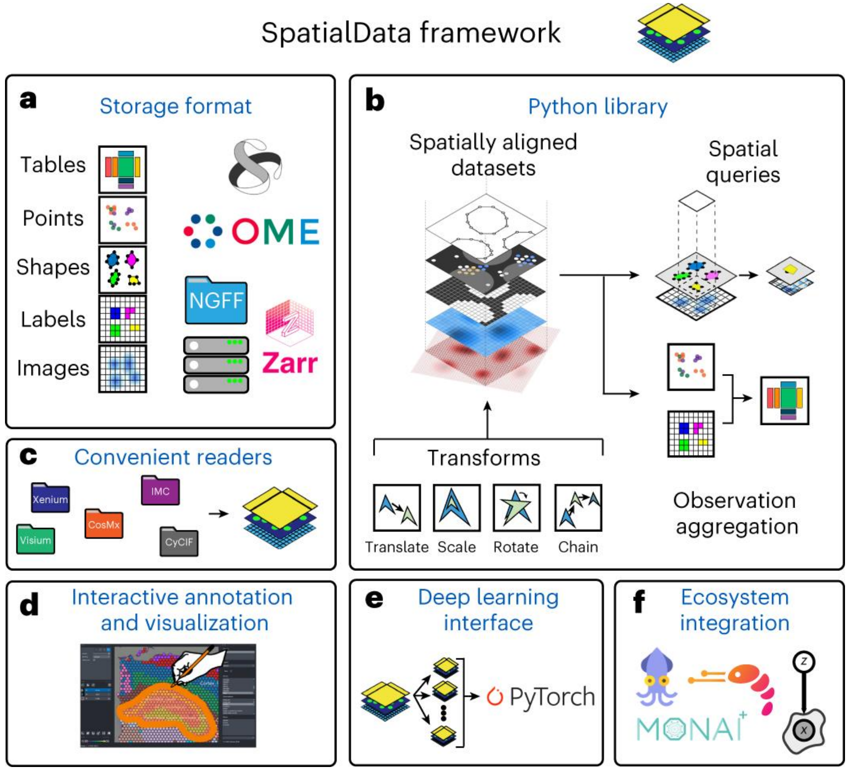
- Run it within SpatialData framework



SpatialData

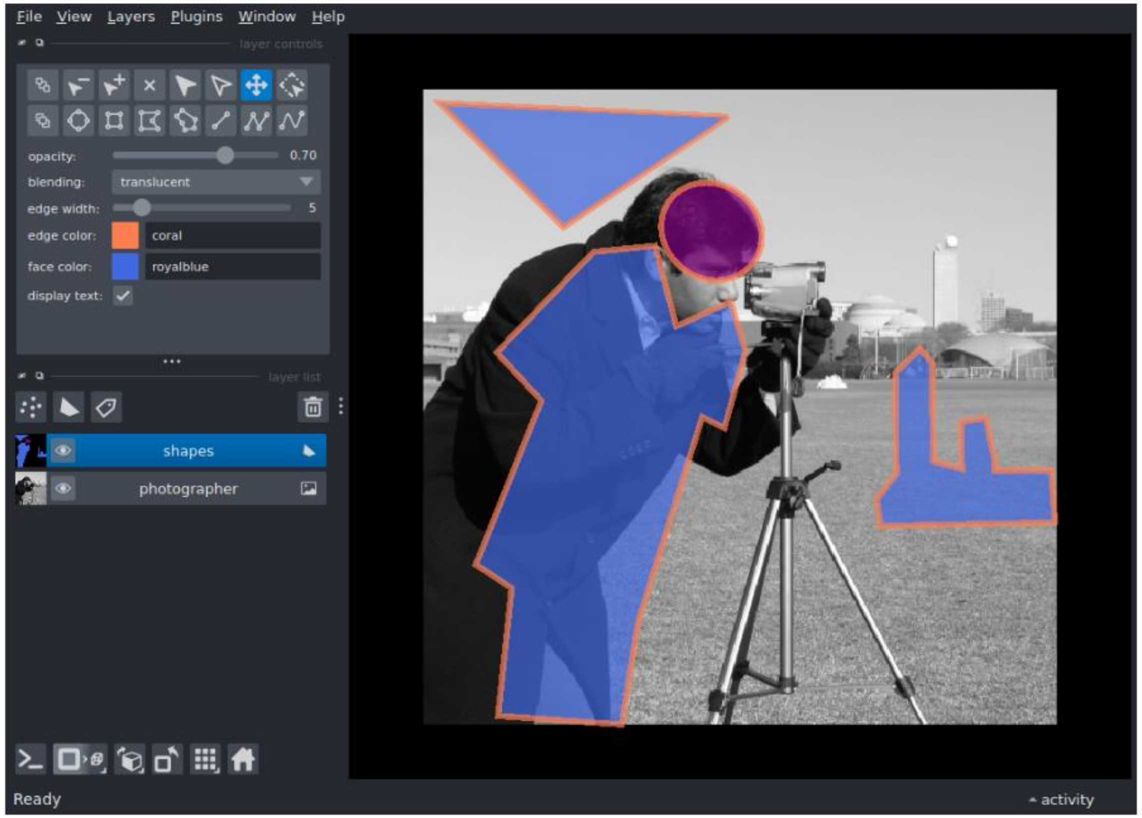
SpatialData framework

```
SpatialData object, with associated Zarr store:
/Users/macbook/embl/projects/basel/spatialdata-sandbox/mouse_liver/data.zarr
├── Images
│     └── 'raw_image': DataTree[cyx] (1, 6432, 6432), (1, 1608, 1608)
├── Labels
│     └── 'segmentation_mask': DataArray[yx] (6432, 6432)
├── Points
│     └── 'transcripts': DataFrame with shape: (<Delayed>, 3) (2D points)
├── Shapes
│     └── 'nucleus_boundaries': GeoDataFrame shape: (3375, 1) (2D shapes)
└── Tables
      └── 'table': AnnData (3375, 99)
with coordinate systems:
    ▸ 'global', with elements:
        raw_image (Images), segmentation_mask (Labels), transcripts (Points),
nucleus_boundaries (Shapes)
```



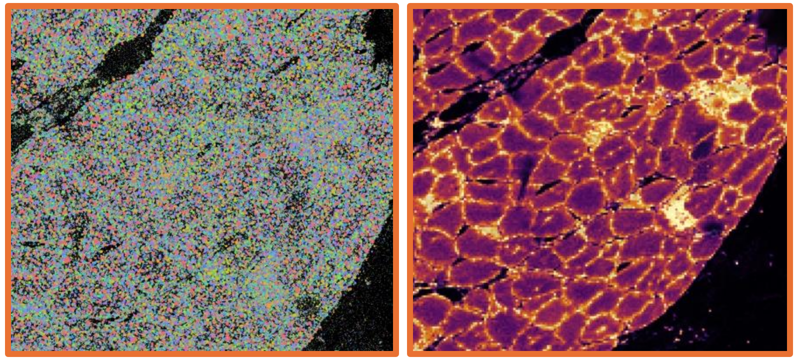
Napari interactive

- Interactive interface to visualize, edit, and add cells manually



Napari: https://napari.org/stable/gallery.html#gallery

- **Transcript-based**



Independent of imaging; works for transcript-rich data

Harder with sparse transcripts or dense tissue

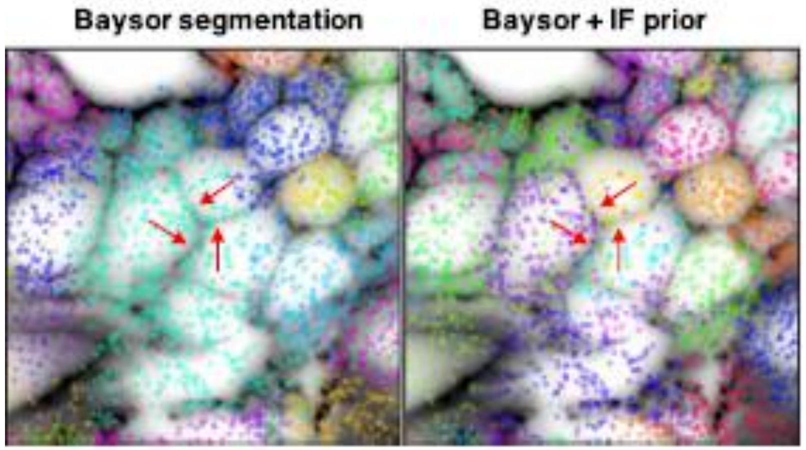
Libraries **Baysor** **Bering** **FICTURE**

- **Combined methods**

Libraries **Segger** **Proseg** **Baysor**

Baysor

- Segmentation-free approach
- Uses molecules / joint likelihood of transcriptional composition
- Can use staining as prior



Segger

- Preprint
- Graph Neural Network (GNN)-based method
- Integrates transcript colocalization with staining

Conclusions

- Segmentation: typically, first step in ST data analysis
- Segmentation matters: effects all downstream analysis
- Segmentation needs might differ per platform and tissue type
- Variety of image-based, transcript-based or combined segmentation methods
- Keep an eye out for benchmarking papers to guide your segmentation approach

Resources

Spatialdata: https://spatialdata.scverse.org/en/stable/
Cellpose: https://www.nature.com/articles/s41592-020-01018-x
Sparrow: https://sparrow-pipeline.readthedocs.io/en/latest/index.html
Sparrow tutorials: https://github.com/vibspatial/targeted_transcriptomics_training
Baysor: https://www.nature.com/articles/s41587-021-01044-w
Segger: https://elihei2.github.io/segger_dev/